

CO2

ASSESSMENT TOOL-1

NAME : P. Ranjithkumar

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REG.NO: 192411119

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1. Use Case Diagram – Hospital Management System

a) Actors Involved

The main actors are:

1. **Patient** – Books appointments and views medical records.
2. **Doctor** – Views patient details and updates medical records.
3. **Administrator** – Manages doctors, patients, appointments, and hospital resources.
4. **Receptionist** – Registers patients and manages appointments.
5. **Pharmacist** – Manages medicines and prescriptions.
6. **Billing Staff** – Manages billing and payments.

b) Primary Use Cases

- Register Patient
- Login
- Book Appointment
- Cancel Appointment
- View Appointment
- Consult Patient
- Update Medical Record
- View Medical Record
- Manage Doctors
- Manage Patients
- Manage Hospital Resources

- Manage Medicines
- Generate Prescription
- Generate Bill
- Make Payment

c) Relationships

Include Relationships

- Book Appointment <<include>> Check Doctor Availability
- Book Appointment <<include>> Confirm Appointment
- Generate Bill <<include>> Calculate Charges
- Make Payment <<include>> Generate Receipt

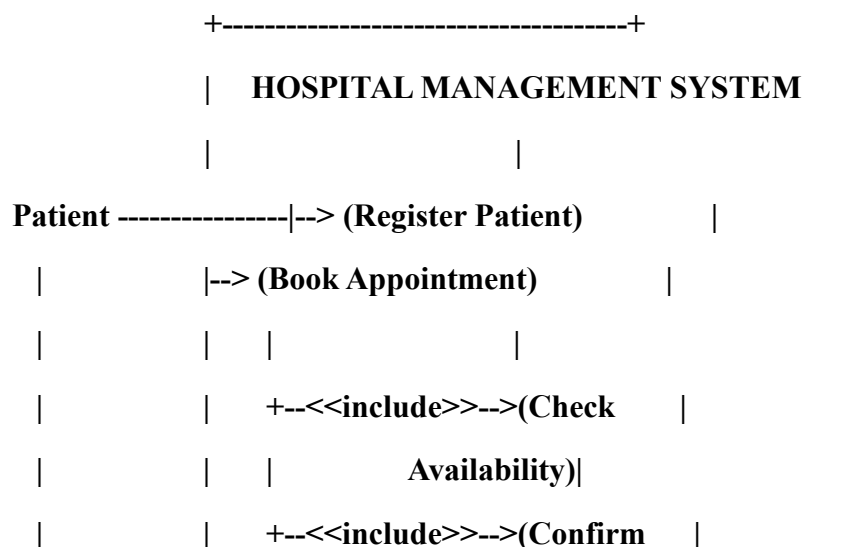
Extend Relationships

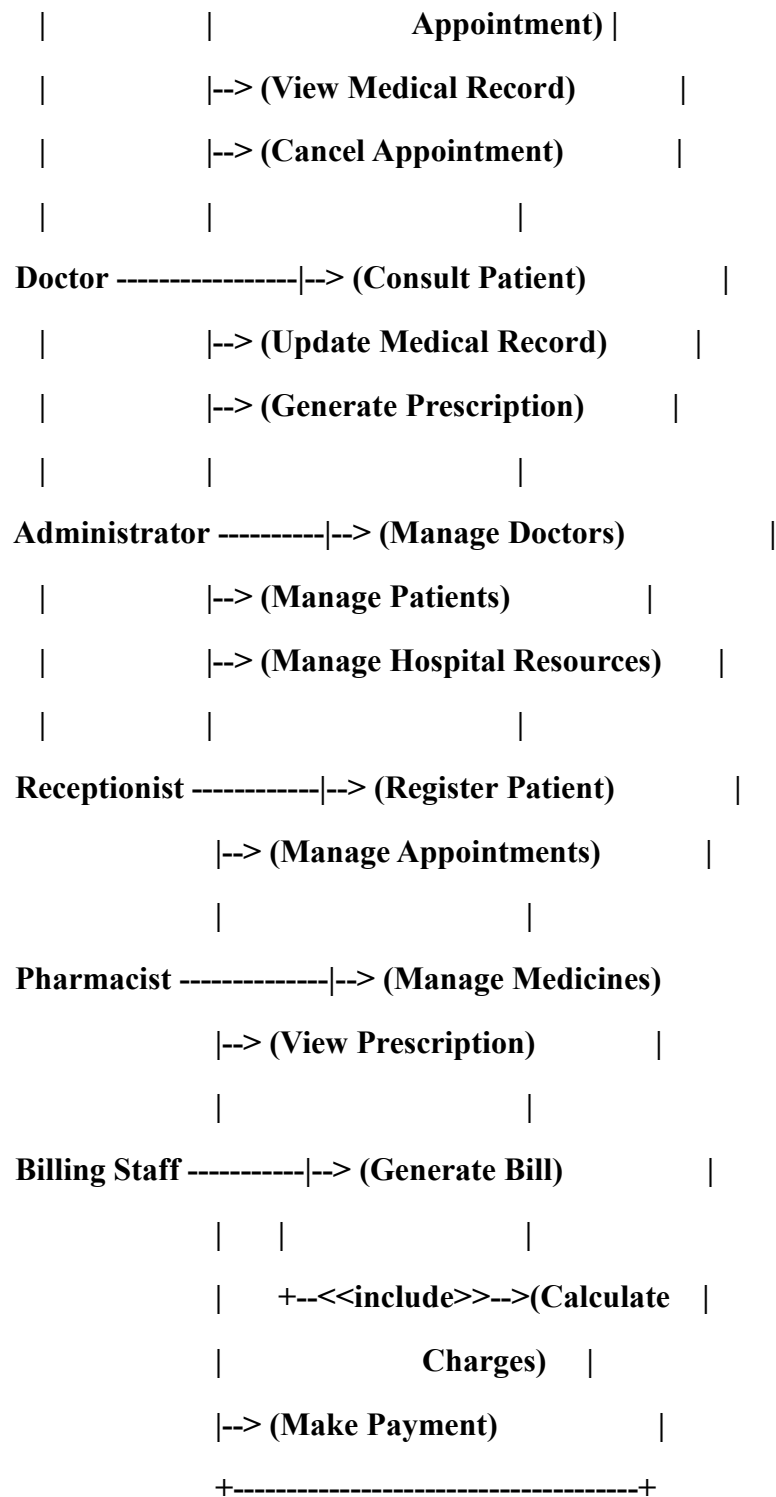
- Cancel Appointment <<extend>> Manage Appointment
- Generate Prescription <<extend>> Consult Patient
- Emergency Treatment <<extend>> Consult Patient

Generalization

- Hospital Staff is a generalized actor.
- Doctor, Receptionist, Pharmacist, Billing Staff, Administrator are specialized actors of Hospital Staff.

d) Use Case Diagram





Conclusion

The Use Case Diagram represents the functional requirements of the Hospital Management System and shows how patients, doctors, administrators, and other hospital staff interact with the system.

2. Class Diagram – Library Management System

a) Classes, Attributes and Methods

1. Book

Attributes:

- bookId
- title
- author
- ISBN
- category
- availabilityStatus

Methods:

- addBook()
- removeBook()
- searchBook()
- checkAvailability()

2. Member

Attributes:

- memberId
- name
- email
- phone

Methods:

- registerMember()
- searchBook()
- borrowBook()
- returnBook()

3. Librarian

Attributes:

- librarianId
- name
- email

Methods:

- addBook()
- removeBook()
- issueBook()
- returnBook()

4. IssueRecord

Attributes:

- issueId
- issueDate
- dueDate
- returnDate
- fine

Methods:

- createIssueRecord()
- calculateFine()
- closeIssueRecord()

5. Library

Attributes:

- libraryId
- libraryName

Methods:

- manageBooks()
- manageMembers()

b) Relationships

Association

- Member **borrow**s Book.
- Librarian **issue**s Book.
- Member **has** IssueRecord.

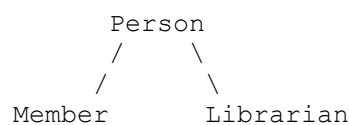
Aggregation

- Library ◇—— Book
- Library ◇—— Member

A Library contains multiple books and members.

Inheritance

A general class **Person** can be created.

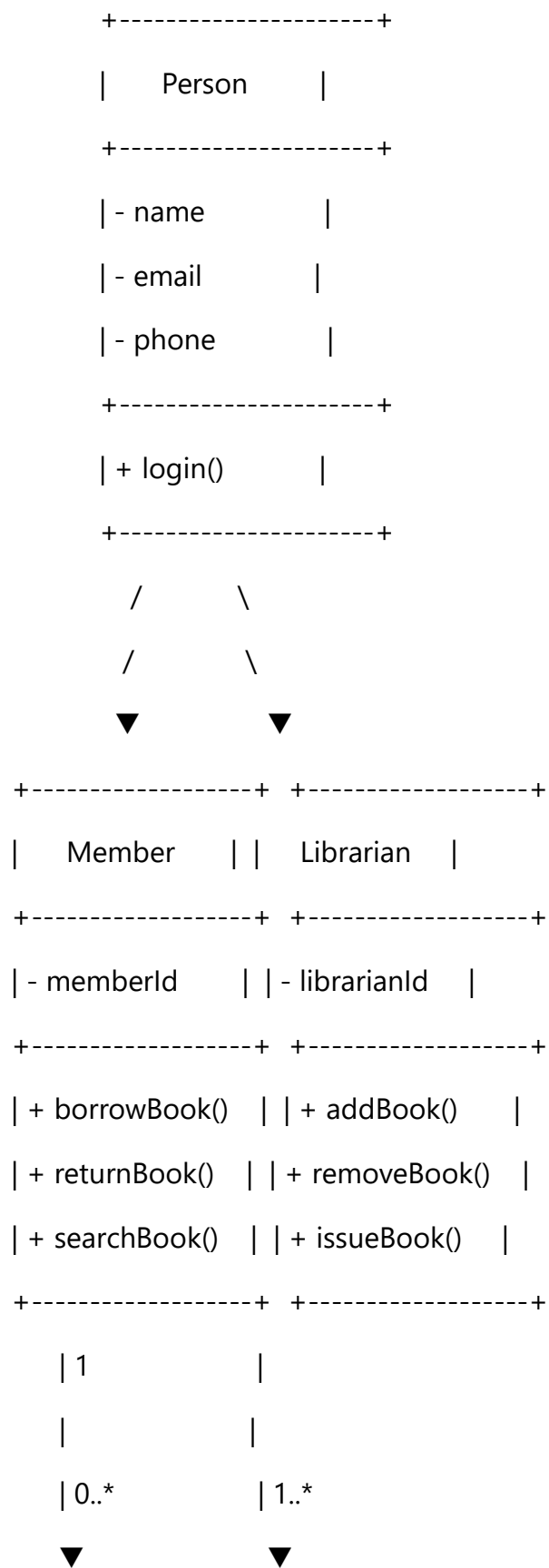


Both Member and Librarian inherit common attributes such as name, email, and phone.

c) Multiplicity

- One Library → Many Books: 1 : 0..*
- One Library → Many Members: 1 : 0..*
- One Member → Many IssueRecords: 1 : 0..*
- One Book → Many IssueRecords: 1 : 0..*
- One Librarian → Many IssueRecords: 1 : 0..*

d) Class Diagram



+-----+

| IssueRecord |

+-----+

| - issuelId |

| - issueDate |

| - dueDate |

| - returnDate |

| - fine |

+-----+

| + createIssueRecord() |

| + calculateFine() |

| + closeIssueRecord() |

+-----+

|

| 0..*

|



+-----+

| Book |

+-----+

| - bookId |

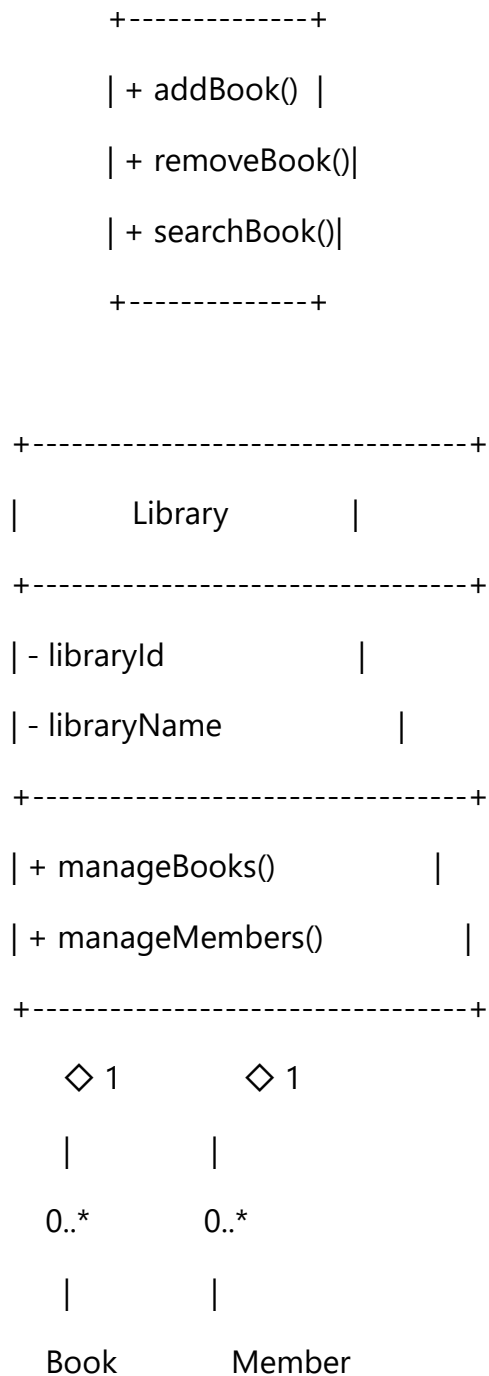
| - title |

| - author |

| - ISBN |

| - category |

| - status |



Conclusion

The Class Diagram shows the static structure of the Library Management System, including classes, attributes, operations, inheritance, aggregation, associations, and multiplicities.

3. Sequence Diagram – Online Food Ordering System

a) Objects Involved

1. Customer
2. Food Ordering System
3. Menu
4. Order
5. Payment Gateway
6. Restaurant
7. Notification Service

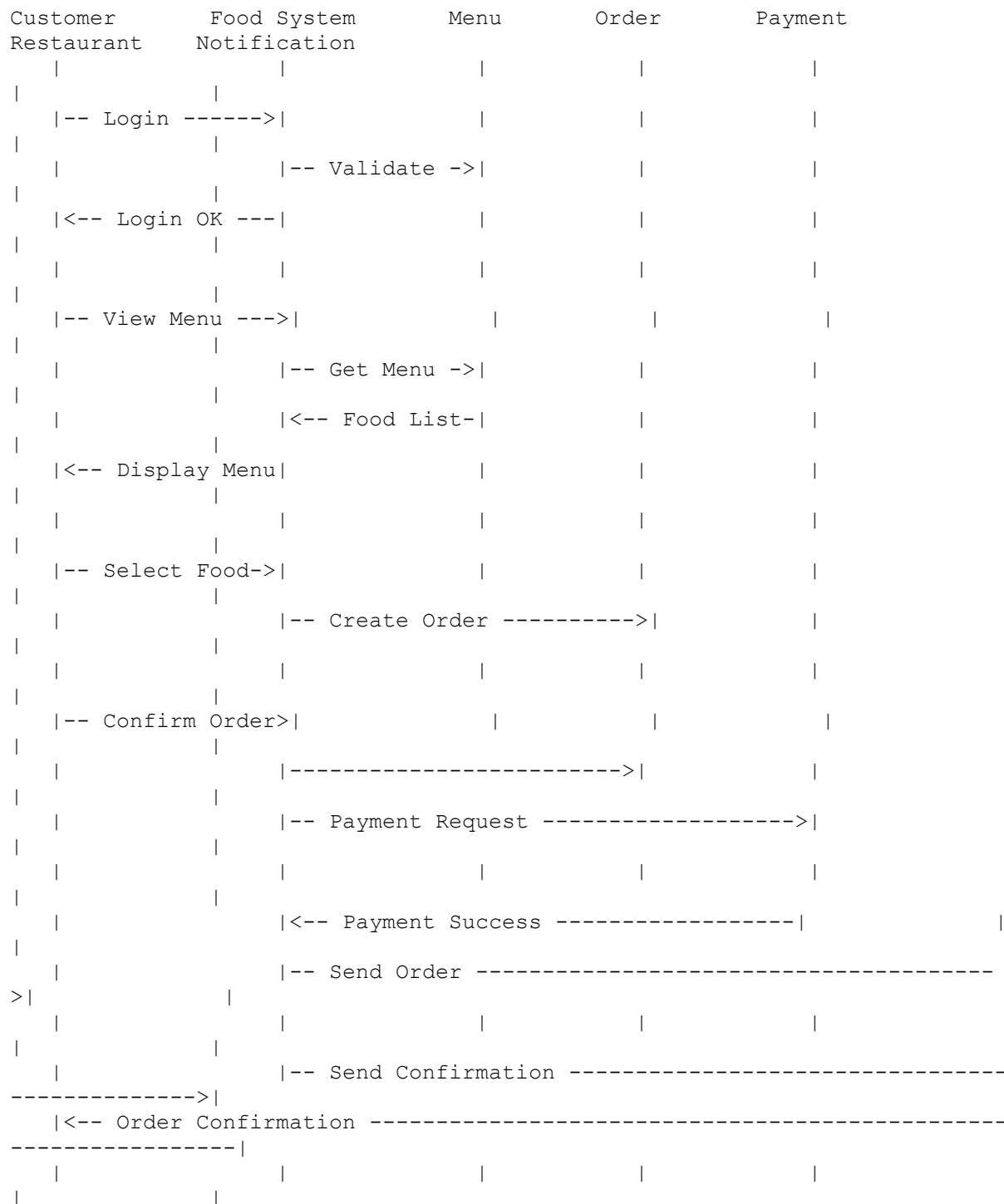
b) Message Flow

1. Customer logs into the system.
2. Customer views food menu.
3. System displays menu.
4. Customer selects food items.
5. System creates order.
6. Customer confirms order.
7. System sends payment request.
8. Payment Gateway processes payment.
9. Payment confirmation is sent to the system.
10. Restaurant receives order.
11. Notification Service sends confirmation to customer.

c) Activation and Lifelines

- **Lifeline:** Vertical dashed line below each object.
- **Activation:** Rectangular box showing when an object is actively processing a request.

d) Sequence Diagram



Conclusion

The Sequence Diagram represents the time-based interaction between the customer, food ordering system, payment gateway, restaurant, and notification service.

4. Activity Diagram – Student Admission System

a) Activities

1. Start
2. Fill Application
3. Submit Application
4. Verify Documents
5. Check Eligibility
6. Approve/Reject Application
7. Pay Admission Fee
8. Confirm Payment
9. Generate Enrollment
10. Send Confirmation
11. End

b) Decision Nodes

- Are documents valid?
- Is the student eligible?
- Is payment successful?

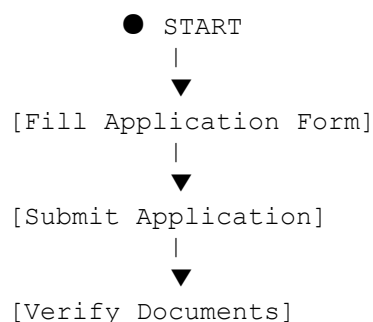
c) Parallel Processing

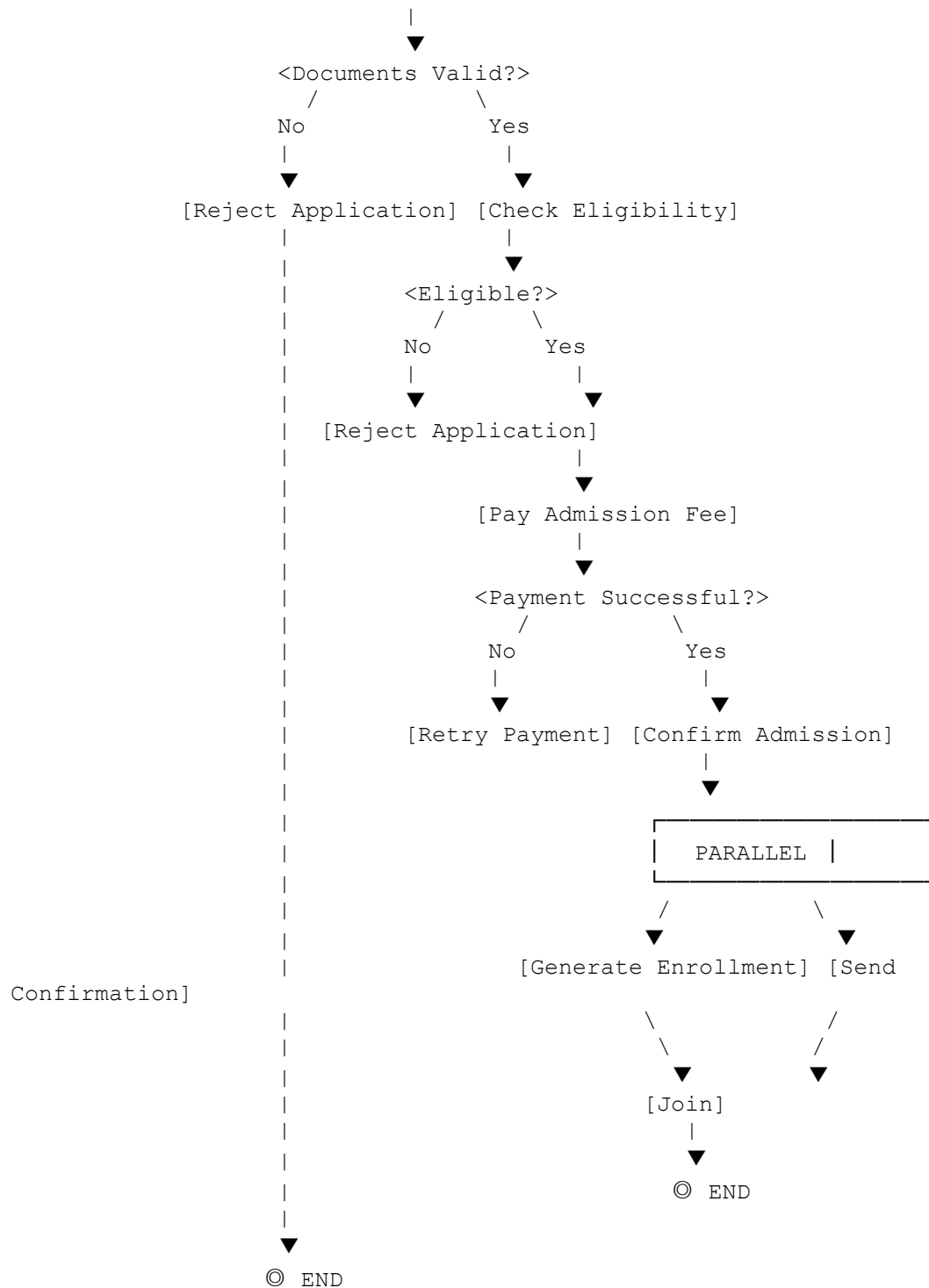
After admission approval, the system can perform:

- Generate enrollment record
- Send confirmation notification

These activities can execute in parallel.

d) Activity Diagram





Conclusion

The Activity Diagram represents the complete admission workflow. Decision nodes handle rejection and payment failures, while parallel activities allow enrollment generation and confirmation notification to occur simultaneously.

5. State Diagram – Vending Machine System

a) Possible States

1. Idle
2. Product Selected
3. Payment Pending
4. Payment Received
5. Product Dispensed
6. Transaction Completed
7. Refund
8. Out of Stock

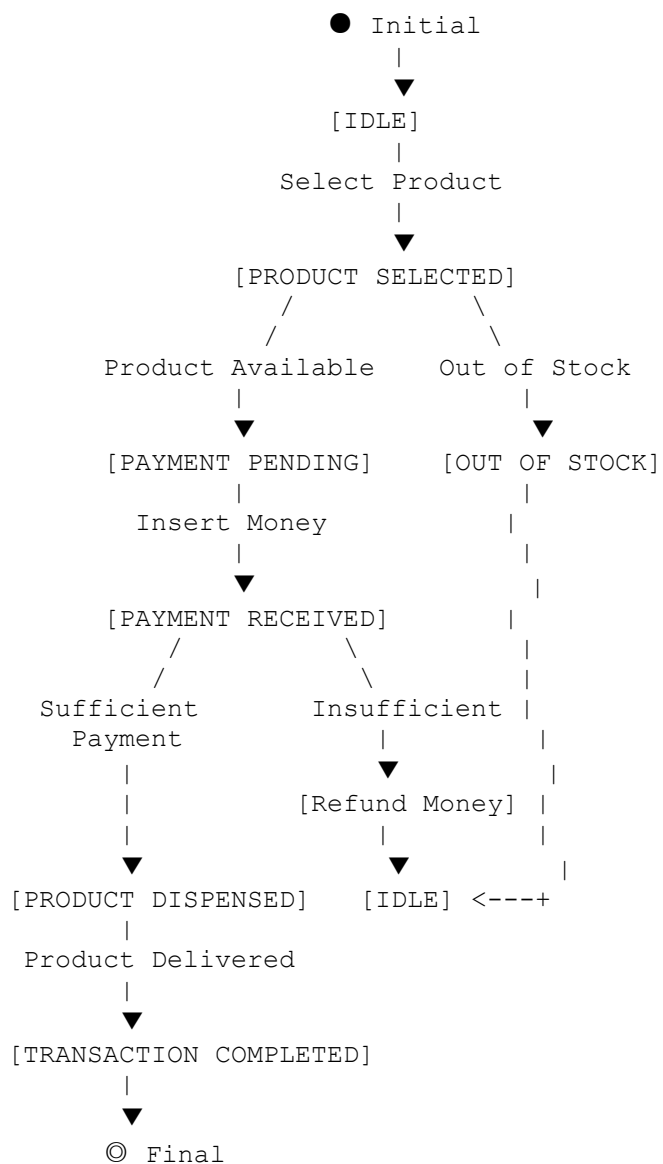
b) Events Triggering Transitions

1. Select product
2. Product unavailable
3. Insert money
4. Payment sufficient
5. Payment insufficient
6. Dispense product
7. Product delivered
8. Cancel transaction

c) Initial and Final States

- Initial State → **Idle**
- Final State → **Transaction Completed**

d) State Transition Diagram



Conclusion

The State Diagram represents how the vending machine changes from one state to another based on user actions and system events.

6. Component Diagram – Online Learning Platform

a) Major Components

1. User Management
2. Course Management
3. Assessment Module
4. Content Delivery Module
5. Notification Service
6. Database
7. Authentication Service

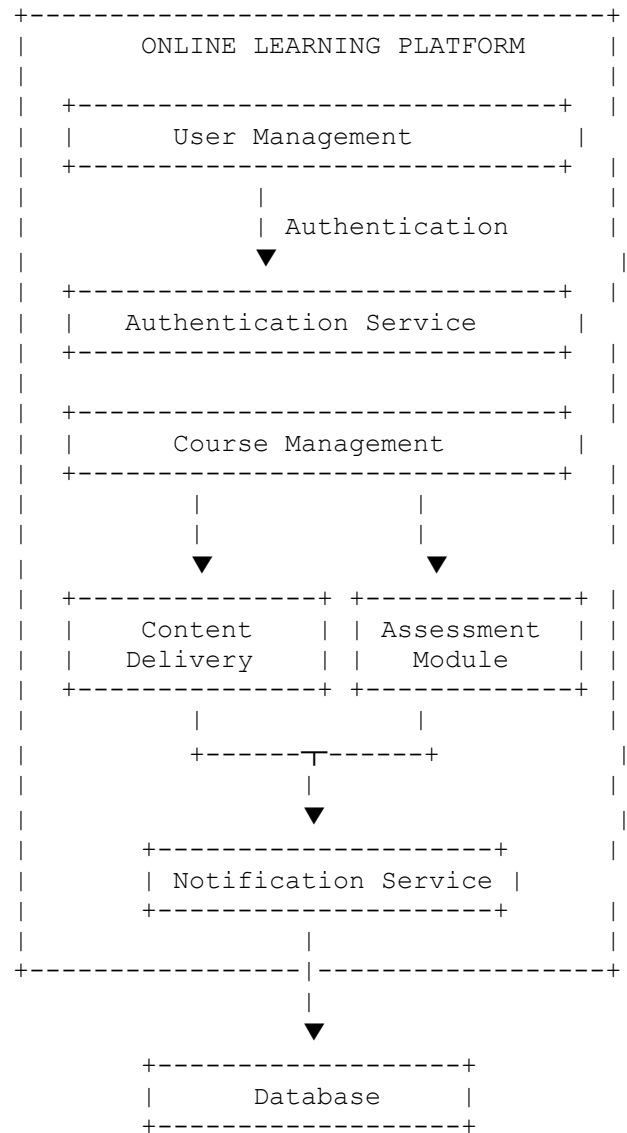
b) Interfaces

- User Management → Authentication Interface
- Course Management → Course Management Interface
- Assessment → Assessment Interface
- Content Delivery → Content Interface
- Notification → Notification Interface

c) Dependencies

- User Management depends on Authentication Service.
- Course Management depends on Database.
- Assessment Module depends on Course Management.
- Content Delivery depends on Course Management.
- Notification Service depends on User Management and Course Management.
- All major modules depend on Database for data storage.

d) Component Diagram



Conclusion

The Component Diagram shows the modular architecture of the Online Learning Platform and illustrates dependencies and communication between different software components.

7. Deployment Diagram – Smart Traffic Monitoring System

a) Hardware Nodes

1. **Roadside Sensors**
2. **Edge/Gateway Device**
3. **Network/Internet**
4. **Cloud Server**
5. **Database Server**
6. **Traffic Authority Computer**
7. **Mobile Alert Device**
8. **Display/Alert System**

b) Software Components

Roadside Sensor

- Vehicle Detection Software
- Traffic Data Collection Software

Edge Gateway

- Data Processing Software
- Data Transmission Service

Cloud Server

- Traffic Monitoring Application
- Violation Detection Module
- Accident Detection Module
- Alert Generation Module

Database Server

- Traffic Database
- Vehicle Data
- Violation Records

Traffic Authority Computer

- Web Browser
- Traffic Monitoring Dashboard

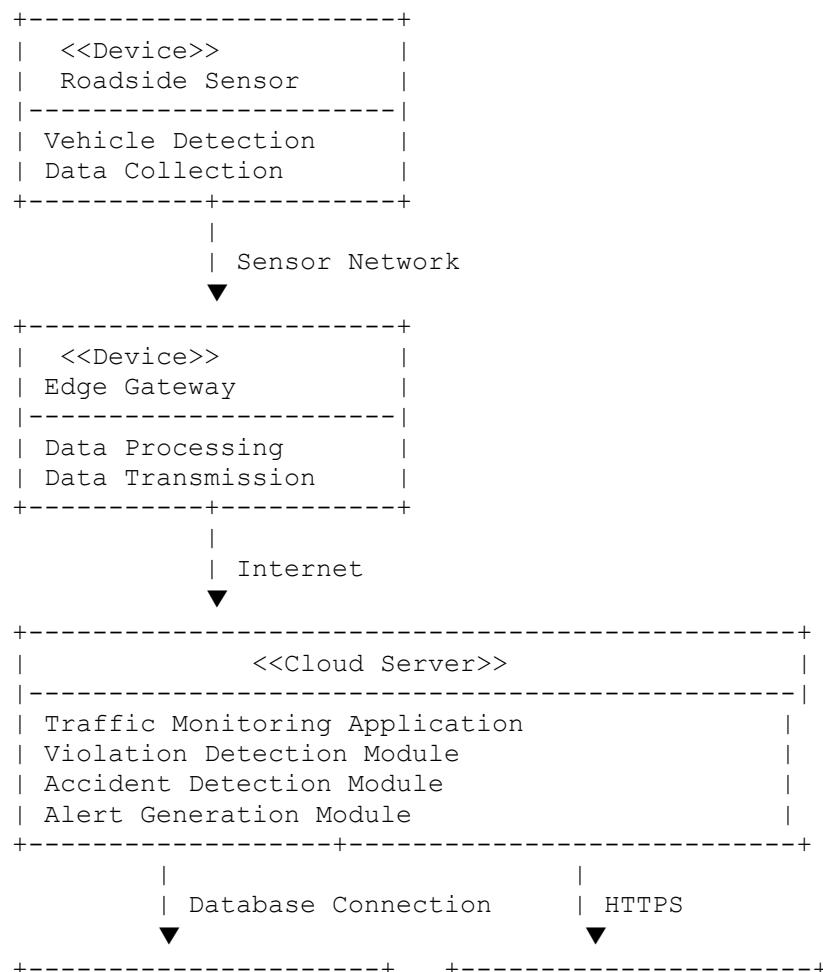
Mobile Device

- Alert Notification Application

c) Communication Links

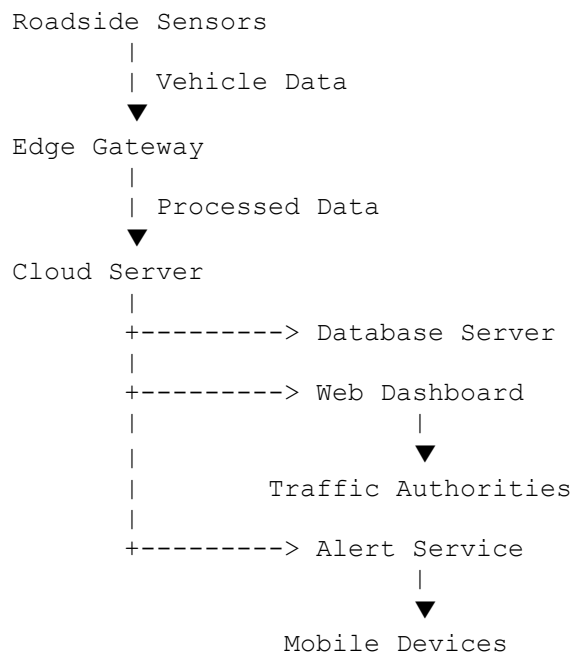
- Roadside Sensors → Edge Gateway: Sensor Network
- Edge Gateway → Cloud Server: Internet
- Cloud Server → Database Server: Secure Network
- Cloud Server → Traffic Authority Dashboard: HTTPS
- Cloud Server → Mobile Devices: Notification Network

d) Deployment Diagram





Complete Communication Flow



Conclusion

The Deployment Diagram shows the physical architecture of the Smart Traffic Monitoring System. Roadside sensors collect vehicle data, the edge gateway processes and transmits the data, and the cloud server analyzes it. Traffic authorities monitor the information through a web dashboard, while violation and accident alerts are sent to authorized devices.

